MIKHIL GEORGE EDATTEL

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EDUCATION

University of Notre Dame *August 2024 - May 2028*

B.S. in Mechanical Engineering ***GPA:*** *3.73/4.00*

Concentration: *Computational Engineering, Thermal & Fluid Sciences*

Relevant Coursework: Thermodynamics, Solid Mechanics, Computational Methods, Fluid Mechanics, Differential Eqns

Honors: President's Circle Mentorship Scholar · Dean's List

University of Notre Dame London - *Summer Engineering Study Abroad* *May 2025 – June 2025*

EXPERIENCE

Cooper Standard | *CAE Engineer Intern* *June 2026 – August 2026*

- Reduced Ford F-150 seal compression force from ~35 to 10 N through nonlinear FEA geometry optimization under ±2 mm door variation and +2.5 mm overslam
- Recommended 0.95 mm Toyota frunk seal flange after analyzing insertion/extraction forces and deformation across 3 designs in MSC Marc/Mentat
- Mitigated GM Silverado glass-run seal buckling through geometry optimization and 3D nonlinear contact FEA of Santoprene and Celloprene

Domer Rover - Mars Rover Design Team | *Project Manager* *September 2024 – Present*

- Lead a 30+ member multidisciplinary rover team, coordinating design reviews and subsystem integration
- Doubled chassis load capacity from 35 to 70 lb. by redesigning the rover's central housing unit (CHU) with 6061 aluminum while allowing for interchangeable arm and astrobiology modules
- Eliminated front-wheel lift by redesigning rocker-bogie suspension mounts guided by center-of-mass analysis
- Increased ground clearance 25% by optimizing CHU height to limit gearbox stress while preserving drill reach

RESEARCH & PROJECTS

Hypersonic Systems Lab | *Research Assistant* *August 2026 – Present*

- Developing a MATLAB/C++ finite-volume CFD solver for compressible flow and shock-wave simulations
- Studying numerical discretization, flux formulations, and shock-capturing methods for flow simulations

H.B Chang Lab | *Research Assistant* *August 2025 – May 2026*

- Miniaturized 2 × 2 × 2 ft nano-fiber manufacturing system into a 500 g handheld prototype by integrating compact mechanical and solution-feeding systems
- Stabilized spinneret at 10,000 RPM by improving coupling geometry in SolidWorks, validated over 20+ tests
- Increased fiber collection efficiency from 85% to 95% with custom-designed 6-channel annular air-jet system

Servo-Controlled Marble Maze *March 2026 – May 2026*

- Designed two-axis gimbal maze in SolidWorks, validating independent ±20° tilt through kinematic analysis and prototype testing
- Fabricated dual-servo prototype with laser-cut and 3D-printed components and produced GD&T drawings

SKILLS

Simulation:	MSC Marc/Mentat · Simulink · Ansys Fluent · MATLAB · Nonlinear Contact Analysis
CAD & Design:	SolidWorks (CSWA) · Siemens NX · Engineering Drawing · GD&T · Dimensional Tolerancing
Programming:	C/C++ · Python · Arduino · Numerical Methods
Manufacturing:	3D Printing · CNC · Lathe · Laser Cutting · MIG Welding · Waterjet Cutting · Soldering